MD. HASAN EMAM

Dhaka, Bangladesh

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Professional Summary

Computer Science and Engineering student at North South University and co-author of a published IEEE Access paper on medical image retrieval, with experience in full-stack web development, machine learning, and deep learning applications. Skilled in PHP, MySQL, Python, Django, and REST API development with hands-on experience building scalable academic and real-world projects. Interested in Artificial Intelligence, Data Science, and Software Engineering with strong problem-solving and project development skills.

Education

North South University **Jan. 2022 – May 2026**
Bachelor of Science in Computer Science and Engineering Dhaka, Bangladesh
CGPA: 3.25/4.00

Haji Lalmia City University College **2019**
Higher Secondary Certificate (HSC), Science Group Gopalganj, Bangladesh
GPA: 4.92/5.00

S.M. Model Govt. High School **2017**
Secondary School Certificate (SSC), Science Group Gopalganj, Bangladesh
GPA: 4.77/5.00

Publications

- M. Saad, **M. H. Emam**, M. I. Sani, and S. Momen, "Concept-Enhanced Hypergraph-Guided Medical Image Retrieval Using CUI Semantics," *IEEE Access*, Oct. 2026. [IEEE Xplore]

Relevant Coursework

• Data Structures & Algorithms • Database Systems • Software Engineering • Object-Oriented Programming • Machine Learning • Deep Learning • Data Communication & Network

Projects

Knowledge-Augmented Radiology Diagnosis and Retrieval System **GitHub**

- Developed an end-to-end medical imaging pipeline for concept prediction, radiology image retrieval, and diagnosis text generation.
- Built a multi-stage deep learning architecture using ResNet-50, concept embeddings, and retrieval fusion techniques.
- Developed diagnosis generation using pretrained vision-language models and CNN-GRU based fallback architecture.
- Evaluated performance using micro-F1, mAP, Recall@K, BLEU, and ROUGE-L metrics.
- **Technologies:** Python, PyTorch, Torchvision, CUDA, NumPy, Pandas, Scikit-learn

Eye Disease Detection System – Deep Learning Web Application **GitHub**

- Developed a medical image classification system using Deep Learning to detect multiple eye diseases from retinal images.
- Integrated CNN-based prediction model with a web application for real-time disease prediction.
- Designed an interactive user interface for uploading retinal images and displaying prediction confidence.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Django

Stock Price Prediction Using Machine Learning **GitHub**

- Developed a machine learning-based stock price prediction system using historical stock market data.
- Implemented multiple regression models including SVR, Decision Tree, Random Forest, Gradient Boosting, Bayesian Regression, and KNN.
- Applied GridSearchCV for hyperparameter tuning and model optimization.
- Evaluated models using MAE, MSE, RMSE, and R^2 Score with graphical visualization of actual vs predicted prices.
- **Technologies:** Python, Pandas, Scikit-learn, Matplotlib, NumPy

Research, Problem Solving & Activities

- Conduct academic research in medical imaging and deep learning, covering model development, debugging, experimentation, and scientific paper writing.
- Practiced programming and problem-solving on HackerRank and other online coding platforms.
- Built academic and personal projects in web development, machine learning, and software engineering.

Technical Skills

- **Languages:** Python, C, C++, Java, SQL, PHP
- **Web:** HTML, CSS, JavaScript, Django
- **Database:** MySQL, SQLite
- **ML/AI:** TensorFlow, Keras, OpenCV
- **Tools:** Git, GitHub, VS Code, Jupyter
- **Concepts:** REST API, OOP, Data Structures